ECE302: Probability and Applications

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1 Random Experiments and Foundations of Probability

There are deterministic models that give the same results every time (e.g. Ohm's and Kirchoff's Laws) and random models (Internet traffic, gait analysis) whose results we cannot predict. Random experiments are experiments in which the outcome is unpredictable when repeated under the same conditions

$$S = \{k_1, k_2, k_3, \ldots, k_N\}$$

S is the sample set of all possible outcomes.
k_i for $i = 1, 2, \ldots, N$ are all of the possible outcomes.

Relative frequency represents how often something happens divided by all outcomes.

$$f_k(n) = \frac{N_k(n)}{n}$$

$$\lim_{n \to \infty} f_k(n) = p_k$$

Three Axioms of Probability

- Axiom 1: $0 \leq P[A] \leq 1$
- Axiom 2: $P[S] = 1$
- Axiom 3: If A and B cannot occur simultaneously then $P[A \; or \; B] = P[A] + P[B]$

Some more notes:

- U is the universal set that contains all possible objects and outcomes in a given application.
- Every set contains the empty set – the set with no elements $\{\}$
- If $A \cap B = 0$ then the sets are disjoint, or mutually exclusive.

Sample Spaces

If S is discrete:

$$P[A] = \frac{\text{\# elements in A}}{N}$$

If S is continuous:

$$P[A] = \frac{\text{area of A}}{\text{area of S}}$$

Regardless of mutual exclusivity, we find that:

$$P[A \cup B] = P[A] + P[B] - P[A \cap B]$$

Intro to Conditional Probability

$$P[B|A] = \frac{P[A \cap B]}{P[A]}$$

2 Conditional Probability, Bayes' Rule, and Independence

Conditional probability is a measure of the probability of an event happening given another event having already occurred.

$$P[A|B] = \frac{P[A \cap B]}{P[B]}$$

The low of total probability expresses the total probability of an outcome as a sum of several distinct events.

$$\begin{aligned} P[A] &= P[A|B_1] \cdot P[B_1] + P[A|B_2] \cdot P[B_2] + \ldots + P[A|B_n] \cdot P[B_n] \\ &= \sum_{i=1}^{N} P[A|B_i] \cdot P[B_i] \end{aligned}$$

Bayes' Rule describes probability of an event based on prior knowledge of conditions that might be related to that event.

$$P[A|B] = \frac{P[B|A] \cdot P[A]}{P[B]}$$

Two variables are independent if the occurrence of one does not affect the probability of the other. For two independent variables:

$$P[A \cap B] = P[A] \cdot P[B]$$

Remember that independent does not mean mutually exclusive.

$$\begin{aligned} P[A|B] &= \frac{P[A \cap B]}{P[B]} \\ &= \frac{P[A] \cdot P[B]}{P[B]} \\ &= P[A] \end{aligned}$$

3 Discrete Random Variables and Expected Value

A Bernoulli trial involves performing an experiment once and noting whether a particular event A occurs. We consider the experiment a success if the event occurs, and a failure otherwise. If k is the number of successes in n independent trials then the probability of k is given by the binomial probability law.

$$p_n(k) = \binom{n}{k} p^k (1-p)^{n-k} \text{ where } \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

A random variable X is a function that assigns a real number to each outcome in the sample space of a random experiment. Discrete random variable assumes values from a countable set.

A probability mass function (pmf) of a discrete random variable is defined as:

$$p_X(x) = P[X = x] = P[\{\beta : X(\beta) = x\}] \text{ for } x \in \mathbb{R}$$

An expected value is the average value of a random variable X is defined as:

$$m_x = E[X] = \sum_{x \in S_X} x p_x(x)$$

From the above, we see that:

$$\begin{aligned} &\text{For } Z = ag(X) + bh(X) + c, \\ &E[Z] = aE[g(X)] + bE[h(X)] + c \end{aligned}$$

4 Variance and Common Discrete Distributions

Variance represents a statistical measure of the spread between numbers in a data set, or, how far a set of numbers is spread out from its mean value.

$$\begin{aligned}\sigma_x^2 &= \mathrm{VAR}[X] \\ &= E[(X - m_x)^2] \\ &= E[X^2] - m_x^2\end{aligned}$$

$E[X^2]$ is called the second moment of X, and we say that in general, $E[X^n]$ is the nth moment of X.

Variance and Conditional Probability

If X is a random variable and event B has occured, then the conditional expected value of $X|B$ is defined as:

$$m_{X|B} = \sum_k x_k p_x(x_k|B)$$

Thus the conditional variance of X given B is:

$$\mathrm{VAR}[X|B] = E[(X - m_{X|B})^2|B] = E[X^2|B] - m_{X|B}^2$$

Random Variables

1. The Bernoulli random variable is the value of the indicator function I_A for the event A. We say $X = 1$ is A occurs and 0 otherwise. Binomial random variables is the number of successes in n Bernoulli trials. The variance of the Bernoulli random variable is given by:

   (a) Probability mass function:

   $$\begin{aligned}p_0 &= 1 - p \\ p_1 &= p\end{aligned}$$

   (b) Expected value function:

   $$E[X] = p$$

   (c) Variance function:

   $$\mathrm{VAR}[X] = p(1 - p)$$

2. The binomial random variable is the number of successes in n Bernoulli trials and hence the sum of n independent, identically distributed Bernoulli random variables.

   (a) Probability mass function:

   $$p_k = \binom{n}{k} p^k (1 - p)^{n-k}$$

   (b) Expected value function:

   $$E[X] = np$$

   (c) Variance function:

   $$\mathrm{VAR}[X] = np(1 - p)$$

3. Geometric random variables are the number of failures before the first success in a sequence of independent Bernoulli trials.

   (a) Probability mass function:

   $$p_k = p(1 - p)^k$$

(b) Expected value function:

$$E[X] = \frac{1-p}{p}$$

(c) Variance function:

$$\text{VAR}[X] = \frac{1-p}{p^2}$$

4. Negative binomial random variables are the number of trials until the $n$he success in a sequence of independent Bernoulli trials.

   (a) Probability mass function:

$$p_k = \binom{k-1}{r-1} p^r (1-p)^{k-r}$$

   (b) Expected value function:

$$E[X] = \frac{r}{p}$$

   (c) Variance function:

$$\text{VAR}[X] = \frac{r(1-p)}{p^2}$$

5. Uniform random variables:

   (a) Probability mass function:

$$p_k = \frac{1}{L}$$

   (b) Expected value function:

$$E[X] = \frac{L+1}{2}$$

   (c) Variance function:

$$\text{VAR}[X] = \frac{L^2-1}{12}$$

6. The Poisson random variable is the number of events that occur in one time unit when the time between events is exponentially distributed with a mean $1/\alpha$.

   (a) Probability mass function:

$$p_k = \frac{\alpha^k}{k!} e^{-\alpha}$$

   (b) Expected value:

$$E[X] = \alpha$$

   (c) Variance function:

$$\text{VAR}[X] = \alpha$$

7. Zipf random variable:

   (a) Few outcomes occur frequently but most occur rarely, seen in applications like Huffamn decoding.

   (b) Probability mass function:

$$p_k = \frac{1}{kc_L} \text{ where } c_L = \sum_{j=1}^{L} \frac{1}{j}$$

   (c) Expected value function:

$$E[X] = \frac{L}{c_L}$$

   (d) Variance function:

$$\text{VAR}[X] = \frac{L(L+1)}{2cL} - \frac{L^2}{c_L^2}$$

5 Cumulative Distribution and Probability Density Functions

A cumulative distribution function represents the probability that the random variable X will be equal to or less than x. It satisfies the following probabilities:

1. Always equals between 0 and 1.
$$0 \leq F_X(x) \leq 1$$

2. Limit as x approaches positive infinity.
$$\lim_{x \to \infty} F_X(x) = 1$$

3. Limit as x approaches negative infinity.
$$\lim_{x \to -\infty} F_X(x) = 0$$

There are three different types of random variables: discrete, continuous, and mixed. A mixed random variable requires a CDF that increases continuously over at least one set of x-values.

$$F_X(x) = pF_1(x) + (1-p)F_2(x)$$

A probability density function is one whose value at any given sample in the sample space can be interpreted as providing a relative likelihood that the value of the random variable would equal that sample.

$$f_x(x) = \frac{dF_X(x)}{dx}$$

It satisfies the following properties:

$$\begin{aligned} f_x(x) &\geq 0 \\ P[a \leq X \leq b] &= \int_a^b f_x(x)\ dx \\ F_X(x) &= \int_{-\infty}^{x} f_x(t)\ dt \\ 1 &= \int_{-\infty}^{+\infty} f_x(x)\ dx \end{aligned}$$

6 Conditional CDF/PDF and More Random Variables

Conditional CDFs and PDFs

Conditional CDFs and PDFs are simple extensions of already familiar concepts. Conditional CDF of x given C is defined as:

$$F_X(x|C) = \frac{P[\{X \leq x\} \cap C]}{P[C]} \text{ for } P[C] > 0$$

and conditional PDF of x given C is defined as:

$$f_X(x|C) = \frac{dF_X(x|C)}{dx}$$

The expected value of a random variable X is defined by:

$$E[X] = \int_{-\infty}^{+\infty} t f_X(t)\ dt$$

More Random Variables

1. Uniform random variable:

   (a) Probability distribution function

   $$f_X(x) = \begin{cases} \frac{1}{b-a}, a \leq x \leq b \\ 0, \text{ otherwise} \end{cases}$$

   (b) Cumulative distribution function

   $$F_X(x) = \begin{cases} 0, x < a \\ \frac{x-a}{b-a}, a \leq x \leq b \\ 1, x > b \end{cases}$$

   (c) Expected value

   $$E[X] = \frac{a+b}{2}$$

   (d) Variance

   $$\text{VAR}[X] = \frac{(b-a)^2}{2}$$

2. Exponential random variable:

   (a) Probability distribution function

   $$f_X(x) = \lambda e^{-\lambda x}$$

   (b) Cumulative distribution function

   $$F_X(x) = \begin{cases} 0, x < 0 \\ 1 - e^{-\lambda x}, x \geq 0 \end{cases}$$

   (c) Expected value

   $$E[X] = \frac{1}{\lambda}$$

   (d) Variance

   $$\text{VAR}[X] = \frac{1}{\lambda^2}$$

3. Gaussian (or normal) random variable:

   (a) Probability distribution function

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{-(x-m)^2}{2\sigma^2}}, -\infty < x < \infty$$

   (b) Cumulative distribution function

$$F_X(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\frac{(x-m)^2}{2\sigma^2}} e^{-\frac{t^2}{2}} \, dt$$

   (c) Expected value

$$E[X] = m$$

   (d) Variance

$$\mathrm{VAR}[X] = \sigma^2$$

7 Common Bounds on Probability Equations

Before we get to bound equations, we have two additional random variable to cover. The Gamma random variable is versatile in that it appears in many equations.

$$
\begin{aligned}
f_X(x) &= \frac{\lambda(\lambda x)^{\alpha-1}e^{-\lambda x}}{\Gamma(\alpha)} \\
\Gamma(z) &= \int_0^\infty x^{z-1}e^{-x}\ dx \\
E[X] &= \frac{\alpha}{x} \\
\mathrm{VAR}[X] &= \frac{\alpha}{x^2}
\end{aligned}
$$

The Cauchy random variable looks like a normal distribution, but it is known as a pathological case.

$$f_X(x) = \frac{a/\pi}{x^2 + a^2}$$

The Cauchy random variable is distinct in that its mean and variance do not exist.

Inequality Equations

- Markov inequality

$$P[X \geq a] = \frac{E[X]}{a}$$

- Chebychev inequality

$$P[|X - m| \geq a] = \frac{\sigma_2}{a^2}$$

- Chernoff provides tighter bounds

$$P[X \geq a] \leq e^{-sa}E[e^{sx}],\ \text{provided } s > 0$$

8 Characteristic Functions and Intro to Multivariate Probability

A variable's characteristic function completely defines its probability distribution. Think of it as a Fourier transform on a random variable.

$$\Phi_X(\omega) = E[e^{j\omega X}] = \int_{-\infty}^{+\infty} f_X(x) e^{j\omega X} \, dx$$

The moment theorem states that moments of X are then given by:

$$E[X^n] = \frac{1}{j^n} \frac{d^n}{d\omega^2} \Phi_X(\omega) \bigg|_{\omega=0}$$

If a random variable is nonnegative, then we can use the z-transform:

$$\begin{aligned} G_N(z) &= \sum_{0}^{\infty} p_N(k) z^{-k} \\ E[N] &= G'_N(1) \\ \mathrm{VAR}[X] &= G''_N(1) + G'_N(1) - (G'_N(1))^2 \end{aligned}$$

Multivariate Probability

In general, we can treat multivariate the same way as we did univariate probability.

$$p(x, y) = P[\{X = x\} \cap \{Y = y\}] \triangleq P[X = x, Y = y]$$

To find the probability of any one variable taking on one value and neglecting the value of the other, we can use the equation:

$$p_x(x_j) = P[X = x_j] = \sum_{k=1}^{\infty} p(x_j, y_k)$$

Marginal distribution function is obtained by removing the constraint on one of the variables.

$$F_X = F_{X,Y}(x, \infty)$$

9 Joint CDFs and PDFs

The PDF of two random variables is obtained from their corresponding CDF. If X and Y are jointly continuous, then the PDF can be obtained by differentiation of the CDF.

$$f_X(x) = \frac{d}{dx}\int_{-\infty}^{x}\left\{\int_{-\infty}^{+\infty} f_{X,Y}(x',y')\ dy'\right\}\ dx' = \int_{-\infty}^{+\infty} f_{X,Y}(x,y')\ dy'$$
$$f_Y(y) = \frac{d}{dy}\int_{-\infty}^{y}\left\{\int_{-\infty}^{+\infty} f_{X,Y}(x',y')\ dx'\right\}\ dy' = \int_{-\infty}^{+\infty} f_{X,Y}(x',y)\ dx'$$

Independence of Joint Variables

X and Y are independent random variables if any event A_1 defined in terms of X is independent of any event A_2 defined in terms of Y:

$$P[X \text{ in } A_1, Y \text{ in } A_2] = P[X \text{ in } A_1]P[Y \text{ in } A_2]$$

Discrete random variables X and Y are independent if and only if the joint PMF is equal to the product of the marginal PMFs for x_i and y_j.

If X and Y are independent random variables, then the defined by any pair of functions $g(X)$ and $h(Y)$ are also independent.

10 Advanced Multivariate Probability

The joint moments of two random variables X and Y summarize information about their joint behaviour. The jk-th joint moment of X and Y for continuous variables is defined by:

$$E[X^jY^k] = \iint_{-\infty}^{+\infty} x^j y^k f_{X,Y(x,y)}\ dx\ dy$$

And for discrete variables:

$$E[X^jY^k] = \sum_i \sum_n x_i^k\ y_n^k\ p_{X,Y}(x_i, y_n)$$

If $j = k = 1$, then $E[XY]$ is called the correlation of X and Y. If $E[XY] = 0$, then we say that X and Y are orthogonal, or have no relation. The jk-th central moment of X and Y is defined by:

$$E[(X - E[X])^j (Y - E[Y])^k]$$

Note that if $j = 2, k = 0$, the expression above would give us $\text{VAR}[X]$ and vice versa.

The covariance of X and Y is defined as the $j = k = 1$ central moment.

$$\text{COV}[X, Y] = E[XY] - E[X] \cdot E[Y]$$

and the correlation coefficients are then given by:

$$\rho_{X,Y} = \frac{\text{COV}[X, Y]}{\sigma_x\ \sigma_y}$$

If X and Y are independent, then X and Y are uncorrelated, i.e.

$$\text{COV}[X, Y] = 0$$

Conditional Probabilities and Expectations

The probability that Y is in A given that we know that $X = x$

$$P[Y \text{ in } A | X = x] = \frac{P[Y \text{ in } A, X = x]}{P[X = x]} \text{ for } P[X = x] > 0$$

There are two distinct cases to consider.

Case 1: X is a discrete random variable

The conditional PMF of Y given X is defined as:

$$p_Y(y|x) = P[Y = y, X = x] = \frac{P[X = x, Y = y]}{P[X = x]}$$

If Y is continuous, the conditional CDF and PDF of Y given X are defined as:

$$F_Y(y|x_k) = \frac{P[Y \leq y, X = xk]}{P[X = xk]} \text{ for } P[X = xk] > 0$$
$$f_Y(y|x_k) = \frac{d}{dy} F_Y(y|x_k)$$

If X and Y are independent:

$$P[Y \leq y, X \leq x] = P[Y \leq y]P[X \leq x]$$
$$F_Y(y|x_k) = F_Y(y)$$
$$f_Y(y|x_k) = f_Y(y)$$

Case 2: X is a continuous random variable

$$F_Y(y|x) = \frac{\int_{-\infty}^{y} f_{X,Y}(x, y')\ dy'}{f_X(x)}$$
$$f_Y(y|x) = \frac{d}{dy} F_Y(y|x) = \frac{f_{X,Y}(x, y)}{f_X(x)}$$

If X and Y are independent:

$$\begin{aligned} f_{X,Y}(x, y) &= f_X(x) f_Y(y) \\ f_Y(y|x) &= f_Y(y) \\ f_X(x|y) &= f_X(x) \\ F_Y(y|x) &= F_Y(y) \\ F_X(x|y) &= F_X(x) \end{aligned}$$

The conditional expectation of Y given X is defined as:

$$E[Y|x] = \int_{-\infty}^{+\infty} y\ f_Y(y|x)\ dy$$

and the k-th moment of Y is given by:

$$E[Y^k] = E[E[Y_k|X]]$$

11 Functions of Two Random Variables

If the random variable Z is defined as a function of two random variables $Z = g(X, Y)$, then:

$$F_Z(z) = P[Z \leq z] = P[X \text{ in } R_z] = \iint f_{X,Y}(x', y')\ dx'\ dy'$$

If X and Y are random variables associated with an experiment, and the random variables Z_1 and Z_2 are defined by two functions of $X = (X, Y)$ as $Z_1 = g_1(X)$ and $Z_2 = g_2(X)$. The joint CDF is then given by:

$$F_{F_1,F_2}(z_1, z_2) = P[g_1(X) \leq z_1, g_2(X) \leq z_2]$$

If we let $Z = AX$, where A is an $n \times n$ matrix, then the joint PDF of Z is given by:

$$f_Z(z) = \frac{f_X(A^{-1}z)}{|A|}$$

Pairs of Jointly Gaussian Random Variables

The random variables X and Y are said to be jointly Gaussian if their joint PDF has the form:

$$f_{X,Y}(x, y) = \frac{\exp\left\{\frac{-1}{2(1-\rho_{X,Y}^2)}\left[\left(\frac{x-m_1}{\sigma_1}\right)^2 - 2\rho_{X,Y}\left(\frac{x-m_1}{\sigma_1}\right)\left(\frac{x-m_2}{\sigma_2}\right) + \left(\frac{x-m_2}{\sigma_2}\right)^2\right]\right\}}{2\pi\sigma_1\sigma_2\sqrt{1-\rho_{X,Y}^2}}$$

Marginal PDFs are given by:

$$f_X(x) = \frac{\exp\left\{\frac{-(x-m_1)^2}{2\sigma_1^2}\right\}}{\sqrt{2\pi}\sigma_1}$$

$$f_Y(y) = \frac{\exp\left\{\frac{-(y-m_2)^2}{2\sigma_2^2}\right\}}{\sqrt{2\pi}\sigma_2}$$

The conditional PDF of X given $Y = y$ is given by:

$$f_X(x|y) = \frac{f_{X,Y}(x, y)}{f_Y(y)} = \frac{\exp\left\{\frac{-1}{2(1-\rho_{X,Y}^2)\sigma_1^2}\left[x - \rho_{X,Y}\frac{\sigma_1}{\sigma_2}(y - m_2) + m_1\right]\right\}}{\sqrt{2\pi\sigma_1^2(1-\rho_{X,Y}^2)}}$$

12 Sums of Random Variables and Central Limit Theorem

Sums of Random Variables

If $X_1, X_2, \ldots, X_n$ are random variables, and their sum is given by $S_n = X_1 + X_2 + \ldots + X_n$, then

$$E[X_1 + X_2 + \ldots + X_n] = E[X_1] + E[X_2] + \ldots + E[X_n]$$
$$\text{VAR}[X_1 + X_2 + \ldots + X_n] = \sum_{k=1}^{n} \text{VAR}(X_k) + \sum_{j=1}^{n} \sum_{k=1, j \neq k}^{n} \text{COV}[X_j, X_k]$$

If S_n represents a sum of independent random variables, then the characteristic function is given by:

$$\Phi_{S_n}(\omega) = E[e^{j\omega S_n}] = E[e^{j\omega(X_1 + X_2 + \ldots X_n)}] = \Phi_{X_1}(\omega)\Phi_{X_2}(\omega)\ldots\Phi_{X_n}(\omega)$$

Sample Means and the Law of Large Numbers

Let X be a random variable for which the mean is unknown. If we have n independent, repeated measurements of X (that is, they are independent, identically distributed n random variables with the same PDF as X, then the sample mean of the sequence is used to estimate $E[X]$):

$$M_n = \frac{1}{n} \sum_{j=1}^{n} X_j$$

The Weak Law of Large Numbers: If $X_1, X_2, \ldots, X_n$ represent a sequence of independent, identically distributed random variables with finite mean $E[X] = \mu$, then for $\varepsilon > 0$:

$$\lim_{n \to \infty} P[|M_n - \mu| < \varepsilon] = 1$$

The Strong Law of Large Numbers: If $X_1, X_2, \ldots, X_n$ represent a sequence of independent, identically distributed random variables with finite mean $E[X] = \mu$ and finite variance:

$$P[\lim_{n \to \infty} M_n = \mu] = 1$$

The Central Limit Theorem

Assume $X_1, X_2, \ldots, X_n$ represent a sequence of independent, identically distributed random variables and their sum is given by S_n with finite mean μ and variance σ^2. Let Z_n represent the zero-mean, unit-variance random variable defined by:

$$Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}} = \sqrt{n}\frac{M_n - \mu}{\sigma}$$

Then:

$$\lim_{n \to \infty} P[Z_n \leq z] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{\frac{-x^2}{2}} \, dx$$

The central limit theorem states that as n becomes large, the CDF of a properly normalized S_n approaches that of a Gaussian random variable.